SUMMARY · MK 6 BUSINESS ANALYTICS · WEEK 8

People Analytics: Social Network Analysis

People analytics usually uses individual HR records (role, team, pay, tenure). This week adds relationships through Social Network Analysis (SNA): employees = nodes, relationships = edges. Using a 360-degree feedback trial in a customer service department (34 nodes, 95 nominations), we cover nodes and edges (directed vs undirected), network-level measures (density, average path length, diameter, components), centrality (degree, in/out-degree, closeness, betweenness, eigenvector), the effect of removing nodes, and community detection with modularity and the Leiden algorithm — compared with formal teams. It ends with a Gephi Lite demo.

Material

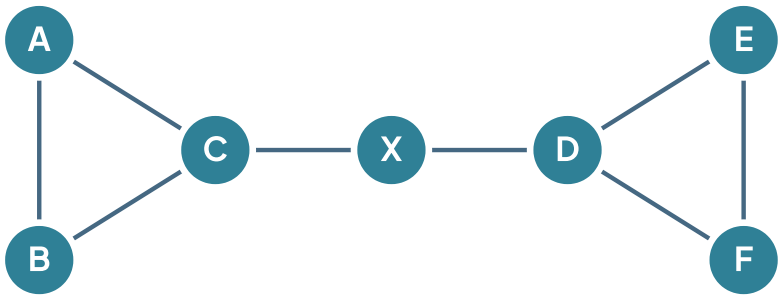
1 Session questions and place in the course

1. What is **social network analysis** and why use it in people analytics?
2. What do **density, average path length and modularity** measure?
3. How is the **centrality** of a node measured?
4. How are **communities** detected and how do they compare with teams?

In the course plan, People Analytics follows Marketing, Financial and Operations Analytics, before Big Data, Deep Learning, Generative AI, Data & AI Governance and the Capstone.

2 What SNA is and why it is used

SNA examines *relationships between individuals*, not only individuals, with measures at three levels: **network, node and community**.



Source	Content
HR records	Describe individuals: role, team, pay, tenure.
Work	Happens through relationships (advice, feedback) that often cross reporting lines.
Questions	Who connects teams and which teams are silos need **relational data**.

Applications: key people (who connects separate teams), new employees (who hasn't been nominated), teams & silos, change agents (whom to inform first), merged units (integrated yet?), collaboration load (who gets most requests).

Org chart

Formal reporting lines.

vs

Network graph

Working relationships: B reports to S1 but is connected to C and D in another team.

3 Nodes and edges: the 360-degree feedback trial

Fictional data: a customer service department ran a one-week 360-degree feedback trial; each employee nominated at least one colleague from any team or level. Participants: **27 department employees + 7 from finance, IT and sales**.

Node list: 34 nodes with role, team, year joined. **Edge list:** 95 nominations.

Edge type	Meaning	Count

Directed	From nominator to nominee	**95**
Undirected	Each pair once	**73**

4-row example

Hendra→Sari, Sari→Hendra, Umar→Hendra, Vina→Umar = 4 directed, 3 undirected edges.

Slide quiz

Ani→Budi, Budi→Ani, Ani→Cici, Dedi→Cici, Cici→Dedi, Budi→Cici: **4 nodes, 6 directed, 4 undirected**.

4 Network-level measures: density, path length, diameter

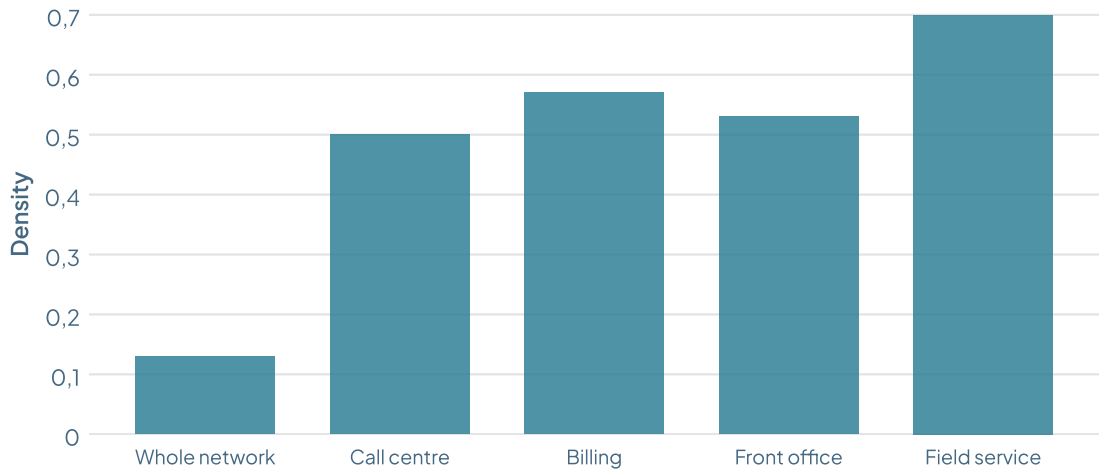
Density = share of possible edges that are present.

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Possible edges = n × (n - 1) ÷ 2 · Density = edges ÷ possible edges
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Example: 7 nodes → 21 possible; 8 present → **0.38**. (Quiz: 5 nodes, 6 edges → **0.60**.)

Graph	Nodes	Edges	Possible	Density
Whole network	34	73	561	**0.13**
Call centre	8	14	28	0.50
Billing	7	12	21	0.57
Front office	6	8	15	0.53
Field service	5	7	10	**0.70**

Density: whole network vs each team



Dense within teams, sparse across the network.

Shortest path: A→F via A, C, X, D, F = 4 edges. **Average path length**: mean shortest path over all pairs = 2.19 (example).

Measure	Value	Meaning
Nodes	34	employees
Undirected edges	73	pairs with ≥1 nomination
Directed edges	95	nominations
Density	0.13	73 of 561
Components	1	all reachable
Avg path length	2.71	mean edges apart
Diameter	6	longest shortest path

5 Centrality: degree, in-degree, out-degree

Degree = a node's number of edges; **degree centrality** = degree ÷ (n – 1). Node C: 3 ÷ 6 = 0.50.

In-degree

Out-degree

Incoming edges = nominations **received**.

Outgoing edges = nominations **made**.

Slide quiz

Highest in-degree: **Cici (3)**; highest out-degree: **Ani and Budi (2)**.

Node	Role	Degree	Centrality	In	Out
Dewi	staff	11	0.333	10	4
Agus	supervisor	10	0.303	9	4
Sari	supervisor	10	0.303	10	4
Maya	supervisor	9	0.273	8	4
Joko	supervisor	8	0.242	6	4
Hendra	manager	7	0.212	7	4

All six have the maximum out-degree 4; they differ in in-degree.

In-degree 0	Team	Joined	Out
Intan	call centre	2026	1
Kevin	call centre	2026	1
Zahra	front office	2026	1
Bayu	field service	2026	1
Iwan	field service	2021	4
Galih	sales	2022	1

HR insight

Four of six joined in 2026 — an onboarding signal. Iwan nominated four but received none.

6 Closeness, betweenness, eigenvector

Measure	Definition	Example	Organisational meaning
Closeness	Reciprocal of average shortest-path distance to all others	X: avg distance 1.67 → **0.60**	Fastest to reach everyone
Betweenness	Share of other pairs' shortest paths passing through the node	X on 9 of 15 pairs → **0.60**	Broker/bridge; a risk point if they leave
Eigenvector	High when neighbours are central	C, D highest **0.45**	Connected to influential people

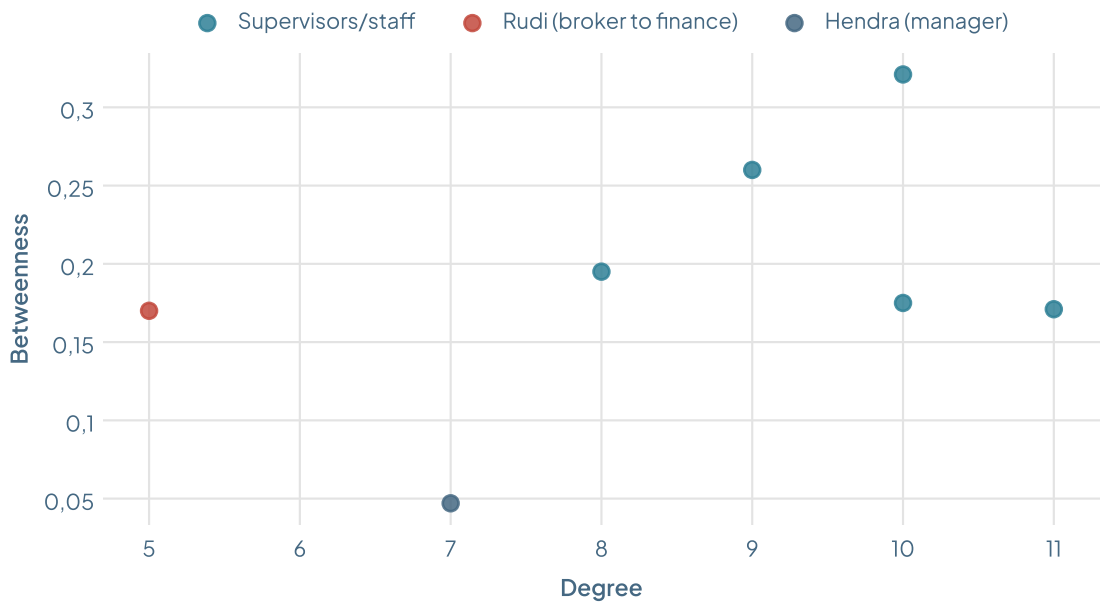
Node	Degree	Closeness	Betweenness	Eigenvector
A, B, E, F	0.33	0.40	0	0.33
C, D	**0.50**	0.55	0.53	**0.45**
X	0.33	**0.60**	**0.60**	0.38

Most important node?

It depends on the measure (slide quiz: P leads degree 0.50 & eigenvector 0.59; Q leads closeness 0.55 & betweenness 0.60).

Measure	1st	2nd	3rd	Hendra
Degree	Dewi 0.333	Agus 0.303	Sari 0.303	0.212, 6th
Closeness	Agus 0.541	Sari 0.516	Maya 0.516	0.493, 5th
Betweenness	Agus 0.321	Maya 0.260	Joko 0.195	0.047, 11th
Eigenvector	Sari 0.385	Dewi 0.340	Agus 0.336	0.327, 4th

Degree vs betweenness centrality (feedback network)



Rudi (degree 5) and Dewi (degree 11) have nearly equal betweenness.

Lesson

High degree does not imply brokerage: Rudi (degree 5) and Dewi (degree 11) have nearly equal betweenness.

7 Removing a high-betweenness node

Removed	Degree	Betweenness	Components	Cut off
Dewi	11	0.171	2	1: Kevin
Sari	10	0.175	2	1: Intan
Hendra	7	0.047	1	0
Rudi	5	0.170	2	**3: Ratna, Sinta, Teguh**

Removing Rudi splits the network and cuts off three finance staff; removing the manager cuts off no one.

Conclusion

The effect of removing a node cannot be inferred from its degree.

8 Community detection and modularity

Community detection partitions nodes into groups with many internal and few external edges, using only **edges**, not team attributes.

Aspect	Modularity
Definition	Share of edges within communities minus the share expected at random with the same degrees.
Reading	0 = no structure; above ~**0.3** = clear structure.
Example	{A, B, C} and {X, D, E, F}: **0.37**.
Leiden	Algorithm finding the partition with the highest modularity.

Slide quiz

Two communities {A, B, C, D} and {E, F, G} (modularity 0.36), linked by one edge D–E.

Leiden finds **five communities (0.52)** in the feedback network vs **0.44** for formal teams — communities describe structure better.

Comm.	Call	Billing	Front	Field	Mgmt	Finance	IT	Sales
1			6		1			2
2	8							
3				5			2	
4		6						
5		1				3		

Rudi

In billing but assigned to community 5 with three finance nodes (3 of his 5 edges). Network data reveal the pattern; explaining it needs follow-up with

Rudi and Agus.

9 Density within and between teams: spotting silos

	Call	Billing	Front	Field
Call centre	**0.50**	0.07	0.06	0.05
Billing	0.07	**0.57**	0.02	0.03
Front office	0.06	0.02	**0.53**	0.03
Field service	0.05	0.03	0.03	**0.70**

Conclusion

Billing, front office and field service share one edge per team pair: they operate as **silos**, connected only through supervisors.

10 Demo: Gephi Lite in the browser

Gephi Lite runs in a browser (gephi.org/gephi-lite) with no installation or programming.

1. Open `feedback-directed.gexf`.
2. Size nodes by **in-degree**.
3. Find nodes with **in-degree 0**.
4. Size nodes by **total degree**.
5. **Detect communities**.

Ethics (extra)

Employee network data is sensitive: use it to improve collaboration and onboarding, not to punish individuals.

Key takeaways

1. SNA studies relationships, not only individuals, at network, node and community levels. Node = employee, edge = relationship.
2. HR data describe individuals; work flows through relationships across reporting lines; silo/connector questions need relational data.
3. Directed edges run nominator → nominee (95); undirected join each pair once (73).
4. Density = edges ÷ [n(n−1)/2]; whole network 0.13, teams 0.50–0.70.
5. Shortest path, average path length (2.71), diameter (6), connected components (1).
6. Degree centrality = degree ÷ (n−1); in-degree = received, out-degree = made; in-degree 0 signals onboarding gaps.
7. Closeness = reciprocal of average distance; betweenness = share of shortest paths through a node (broker); eigenvector = connected to central nodes.
8. The most important node depends on the measure; high degree ≠ broker; removal effects can't be inferred from degree.
9. Community detection uses edges only; modularity above ~0.3 = clear structure; Leiden maximises it (0.52 vs 0.44 for teams).
10. Low between-team density = silos; network data reveal patterns, explanations need follow-up; Gephi Lite demo.

Case studies

Case & worked answers

Session case: reading the 360-degree feedback network (fictional slide data)

Fictional company and people, as stated on the slides. Three recommendations for Hendra: (1) onboarding buddies for the four 2026 joiners with in-degree 0; (2) broker risk — Rudi is the only bridge to finance, so build backup ties and include him in retention plans; (3) silos — cross-team customer-case forums for billing, front office and field service. To spread change fast, start with the highest-closeness nodes (Agus, Sari, Maya), not automatically the manager. Network findings always need confirming through conversation.

Worked examples

Problem 1 **Easy** **Density and degree centrality**

A 6–person network (A–F) with 7 ties: A–B, A–C, B–C, C–D, D–E, E–F, D–F. Compute the network density and C's degree centrality.

1.
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Possible edges = n(n - 1) ÷ 2 = 6 × 5 ÷ 2 = 15
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2. Density = 7 ÷ 15 ≈ **0.47**.

3. C's degree = 3 (A, B, D).
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Degree centrality = degree ÷ (n - 1) = 3 ÷ 5 = 0.60
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Answer: Density 0.47; C's degree centrality = 0.60.

Problem 2 Medium **Closeness and betweenness**

Use the same network. Compute the closeness centrality of A and D, and C's (normalised) betweenness. Who is the "bridge" between groups?

1. Shortest distances from A: B 1, C 1, D 2, E 3, F 3 → total 10.

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Closeness = (n - 1) ÷ Σ distances = 5 ÷ 10 = 0.50
```

2. From D: C 1, E 1, F 1, A 2, B 2 → total 7 → closeness = 5 ÷ 7 ≈ **0.71**.

3. C's betweenness: pairs whose shortest path *must* pass through C = {A, B} × {D, E, F} = 6 pairs. Normalise ÷ [(n – 1)(n – 2) ÷ 2] = 6 ÷ 10 = **0.60**. D is also 0.60 (pairs {A, B, C} × {E, F}).

4. C and D are the bridges: if the C–D tie breaks, the network splits in two. They matter for information flow, but are also a weak point.

Answer: Closeness A 0.50, D 0.71; betweenness C = D = 0.60.

Problem 3 Medium **Within- vs between-team density (silos)**

Same network. Team X = {A, B, C}, team Y = {D, E, F}. Compute the density within each team and between the teams. Is there a silo?

1. Within X: ties A–B, A–C, B–C = 3 of 3 possible → density **1.0**. Within Y: D–E, E–F, D–F = 3 of 3 → **1.0**.
2. Between teams: possible = 3 × 3 = 9; only C–D exists = 1 → density **0.11**.
3. Very dense inside, very sparse between → a **silo** pattern, with all coordination resting on the single C–D tie. Actions: cross-team projects, rotation or shared forums.

Answer: Within teams 1.0; between teams 0.11 → silo.

Practice questions

Try answering first. The answer key is below each question.

1. In SNA for people analytics, a 'node' and an 'edge' are…

A. Team and department
B. An employee and a relationship between two employees
C. Role and pay
D. Manager and formal subordinate

Answer: B. Node = employee; edge = relationship.

2. Why is network analysis needed beyond ordinary HR records?

A. HR records lack pay
B. Work flows through cross-line relationships; connector and silo questions need relational data
C. To replace the org chart
D. It's cheaper than surveys

Answer: B. It adds relationships to people analytics.

3. Ani→Budi, Budi→Ani, Ani→Cici, Dedi→Cici, Cici→Dedi, Budi→Cici. Number of UNDIRECTED edges?

A. 3
B. 4
C. 5
D. 6

Answer: B. Unique pairs = 4 (6 directed).

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4. A 7-node graph with 8 edges has density…

A. 0.13
B. 0.38
C. 0.57
D. 1.14

Answer: B. 21 possible; 8 ÷ 21 = 0.38.

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5. The feedback network has 34 nodes and 73 undirected edges. Density?

A. 0.07
B. 0.13
C. 0.26
D. 0.50

Answer: B. 561 possible; 73 ÷ 561 = 0.13.

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6. A network's diameter is…

A. The average shortest path
B. The longest shortest path among all pairs
C. The number of components
D. The maximum number of edges

Answer: B. Feedback network: average 2.71, diameter 6.

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7. In the directed 360 network, an employee's in-degree is…

A. Nominations made
B. Nominations received
C. Team size
D. Tenure

Answer: B. In = received; out = made.

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8. Four of six employees with in-degree 0 joined in 2026. The most sensible HR reading is…

A. They perform poorly
B. New hires aren't connected yet — an onboarding signal
C. The data is wrong
D. They should be dismissed

Answer: B. Not nominated ≠ poor performance.

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9. Dewi's degree centrality (degree 11) in a 34-node network is…

A. 0.324
B. 0.333
C. 0.500
D. 11

Answer: B. 11 ÷ 33 = 0.333.

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10. Which measure is the share of other pairs' shortest paths passing through a node (brokerage)?

A. Degree
B. Closeness
C. Betweenness
D. Eigenvector

Answer: C. X lies on 9 of 15 pairs → 0.60.

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11. A node scoring high because its neighbours are central is measured by…

A. Eigenvector centrality
B. Out-degree
C. Density
D. Diameter

Answer: A. C and D score 0.45, linked to the triangle and X.

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12. In the example, X has only 2 edges but the top closeness and betweenness. The lesson?

A. X was miscounted
B. The most important node depends on the measure
C. Degree is always best
D. Closeness equals degree

Answer: B. C/D lead degree & eigenvector; X leads closeness & betweenness.

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13. Removing Rudi (degree 5) cuts off 3 finance staff; removing Hendra (degree 7) cuts off none. Conclusion?

A. Degree determines removal impact
B. Removal impact can't be inferred from degree
C. Managers are always most important
D. Finance doesn't matter

Answer: B. Rudi is a broker — key for retention and succession.

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14. Community detection groups nodes based on…

A. Formal team attribute
B. Edges only: many within, few between
C. Year joined
D. Role

Answer: B. It ignores team attributes, so it can be compared with teams.

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15. Modularity 0.52 for Leiden vs 0.44 for formal teams means…

A. Teams describe structure better
B. Leiden communities describe the structure better than formal teams
C. No community structure
D. The network split in two

Answer: B. Higher modularity = better partition; above ~0.3 is clear structure.

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16. Between-team density among billing, front office and field service is only 0.02–0.03. This shows…

A. Strong cross-team collaboration
B. The three teams work as silos, linked via supervisors
C. Teams are too big
D. Incomplete data

Answer: B. High diagonal, very low off-diagonal → silos.

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17. Rudi (billing) is placed in the finance community. The analyst's best next step?

A. Move Rudi to finance
B. Follow up with Rudi and Agus to understand why
C. Drop Rudi from the data
D. Conclude Rudi is disloyal

Answer: B. Network data reveal the pattern; explaining it needs follow-up.

Sources

- P3MD Week 8 MK6 — Business Analytics: People Analytics (Social Network Analysis)

- Gephi Lite (gephi.org/gephi-lite); Leiden algorithm (Traag et al., 2019)